

Tianyang Che

Large-Scale System Architect · Platform/Product Builder (0 -> 1 -> 100) · Big Tech + Startup Operator

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SUMMARY

Seasoned tech lead with 10+ years architecting large-scale distributed systems across Meta, Amazon, Coupang, and a Series A AI supply chain startup. Hands-on experience designing and deploying LLM-powered agentic AI systems, NL→SQL reasoning pipelines, and AI-driven workflow automation. Deep expertise in distributed systems, microservice architecture, and full-stack engineering across cloud platforms. Proven ability to lead cross-functional teams, drive AI adoption across engineering workflows, and deliver measurable impact at scale. Inventor of a granted Korean patent and a pending U.S. patent.

EXPERIENCE

Senior Software Engineer

[Auger, Inc.](#)

May 2025 – Present

Series A AI startup (\$100M, Oak HC/FT), founded by ex-Amazon's Dave Clark — building the first AI-powered end-to-end supply chain OS.

- **Architected and deployed Auger's flagship LLM-powered agentic AI system** end-to-end — designing the NL→SQL reasoning and multi-step tool-execution pipeline with RAG-based context retrieval, enabling secure Snowflake query generation and context-aware response synthesis for automated supply chain decision-making workflows.
- Drove AI adoption across engineering workflows by integrating LLM-based automation into CI/CD-adjacent data pipelines — reducing manual intervention in supply chain analytics and accelerating feedback loops for on-call incident response.
- **0→1 built multi-tenancy data platform** processing billions of supply chain events daily, powering all data-related use cases in flagship products — including agentic chatbot, ML infra, data visualization, reporting and analysis.
- **Tech Lead for 10+ engineer team**, establishing sprint cadence, technical design review, and on-call rotations in a zero-process startup environment.

Staff Software Engineer II

[Coupang, Inc.](#)

Nov 2021 – May 2025

- **Founding engineer of Rocket Growth**, a multi-billion business helping Coupang sellers succeed by handling product packaging, storage, shipping, management and customer service on their behalf.
- **Tech Lead, Supply Chain Management (7 engineers)**: pioneered the design, development and enhancement of the entire Reverse Logistics system from scratch, including customer return and vendor return.
- **Architected the inventory loss and damage compensation system**, reducing compensation costs by 50% — equivalent to approximately 1% of GMV.
- **Tech Lead, Inventory Management**: led design and development of the logistics center storage fee, contributing tens of millions of dollars in revenue. Invented an inventory age calculation algorithm; patent filed and approved.
- Drove technical direction, engineering processes, and best practices across the org.
- Mentored engineers across teams; facilitated promotions of 5 engineers.

Software Engineer

[Meta Platforms, Inc.](#)

Jun 2020 – Nov 2021

- **Zero-regression privacy enforcement at billion-user scale**: modernized three foundational Ads Infra data sources — user profile storage, MySQL ad attribute databases, and Hive table indexing — enforcing privacy constraints with no performance regression.
- **First production customer of Meta's company-wide privacy solution**: pioneered adoption within Ads Infra through cross-org collaboration, establishing a blueprint followed by numerous teams.

Software Development Engineer II

[Amazon.com, Inc.](#)

Mar 2015 – Jun 2020

Amazon Go

- **Cut infrastructure cost by 50%** by re-architecting Amazon Go's failure-resilient store tracking systems, maintaining 99.99%+ uptime globally.
- Built global accuracy monitoring infrastructure to automatically detect and escalate accuracy issues across the Amazon Go store fleet.

Alexa AI

- **Designed Alexa's customer satisfaction metric system** using ML-based logic, providing leadership a quantitative signal on human-Alexa interaction quality across all global traffic.

- Built hourly Spark data pipeline at Alexa scale, publishing metrics to AWS Redshift, CloudWatch, and the A/B testing platform; built Spark SQL tooling for ad hoc query access across the org.

Amazon Prime & Payments Platform

- **Reduced P100 payment query latency by 20%+** by introducing a cache layer on the Payments Core Engine.
- Contributed to Amazon Mexico Retail launch by migrating payment workflows to idempotent designs.
- Contributed to Amazon Key launch — enabling couriers to deliver directly inside customer homes and vehicles.

INDEPENDENT PROJECTS

[Smart Waste Classifier](#) · iOS App

2025 – Present

- **Designed and shipped an AI-powered iOS app** enabling waste classification (recycle / compost / landfill) via camera — owned full product lifecycle from UX design to App Store submission.
- Integrated Apple Vision framework with an LLM-based reasoning pipeline; leveraged OpenAI APIs to improve edge-case accuracy.

TECHNICAL SKILLS

Languages	Java, Python, SQL, JavaScript, TypeScript
AI & ML	LLMs (OpenAI, Azure OpenAI), Agentic AI, RAG, LLM Fine-tuning, NL→SQL, ML Pipelines, A/B
Testing Infrastructure	
Cloud & Infra	AWS (full-stack), Azure (full-stack), Docker, Kubernetes, Terraform, Jenkins, GitLab CI
Frameworks	React, Node.js, Spring, Django/Flask, REST APIs, Microservices
Data & Streaming	Apache Spark, Kafka, Flink, Elasticsearch, Redis, MySQL, PostgreSQL, MongoDB, Temporal

PATENTS

- South Korean Patent KR10-2024-001478 — Novel Inventory Age Calculation Algorithm for Multi-Warehouse Logistics (Granted 2025)
- U.S. Patent Application US18/942068 — Inventory Management Method and System (Pending)

EDUCATION

M.S. Computer Science ·	Illinois Institute of Technology, Chicago IL	GPA 3.9 2012 – 2014
<i>Algorithm Design, Data Mining, Parallel & Distributed Computing</i>		
B.S. Computer Science & Technology ·	Jilin University, China	GPA 3.2 2008 – 2012